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Class: T2 Batch: T21

ASD&D Experiment No. 06

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Objective:
To practically demonstrate Agile project management by creating a Work Breakdown Structure (Epics, Stories, Tasks), assigning a balanced team workload, executing a Sprint, and integrating a continuous DevOps pipeline where GitHub code commits automatically transition Jira issue statuses.

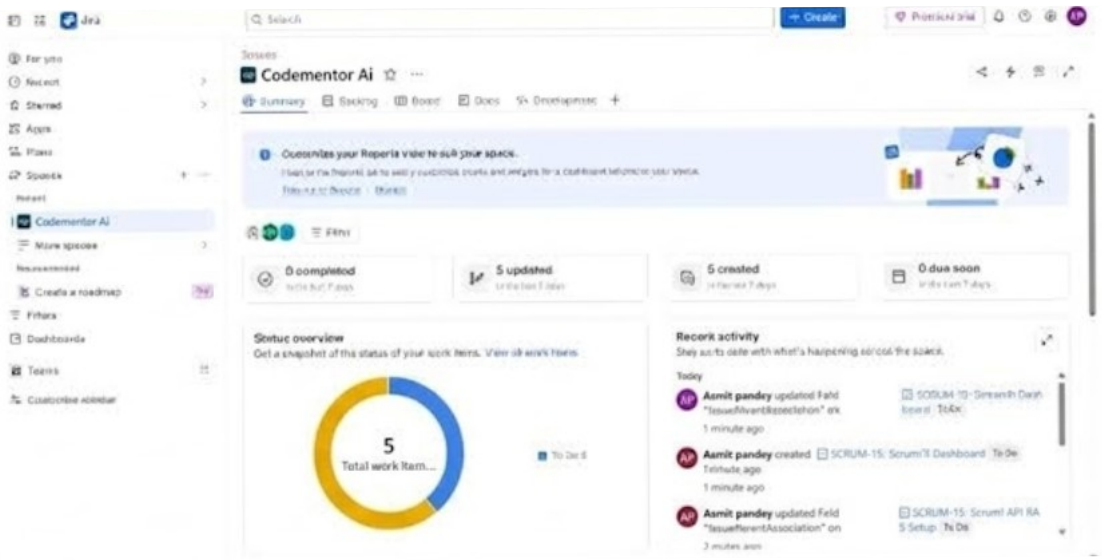
Theory:

- Agile & Scrum: An iterative software development framework. Work is broken into Epics (large initiatives) and User Stories (functional requirements), assigned to team members, and executed in time-boxed cycles called Sprints.
- Jira: A project management platform that tracks issues, manages workflows via Scrum boards, visualizes dependencies, and monitors workload distribution across the team.
- DevOps Integration (Smart Commits): Connecting a version control system (GitHub) to Jira allows developers to update issue statuses remotely by tagging Jira issue keys and transition commands (e.g., #done) directly in their Git commit messages.

Project Name: CodeMentor AI
Project Key: GIT
Methodology: Scrum

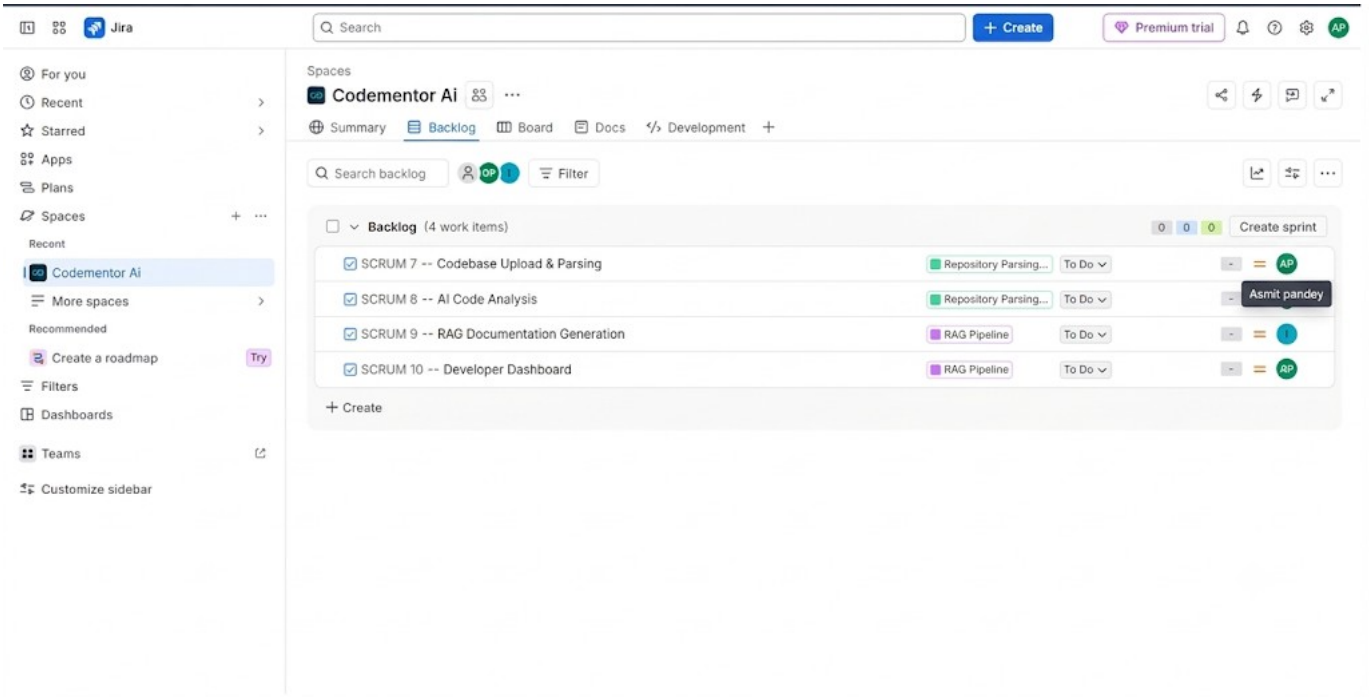
Procedure:

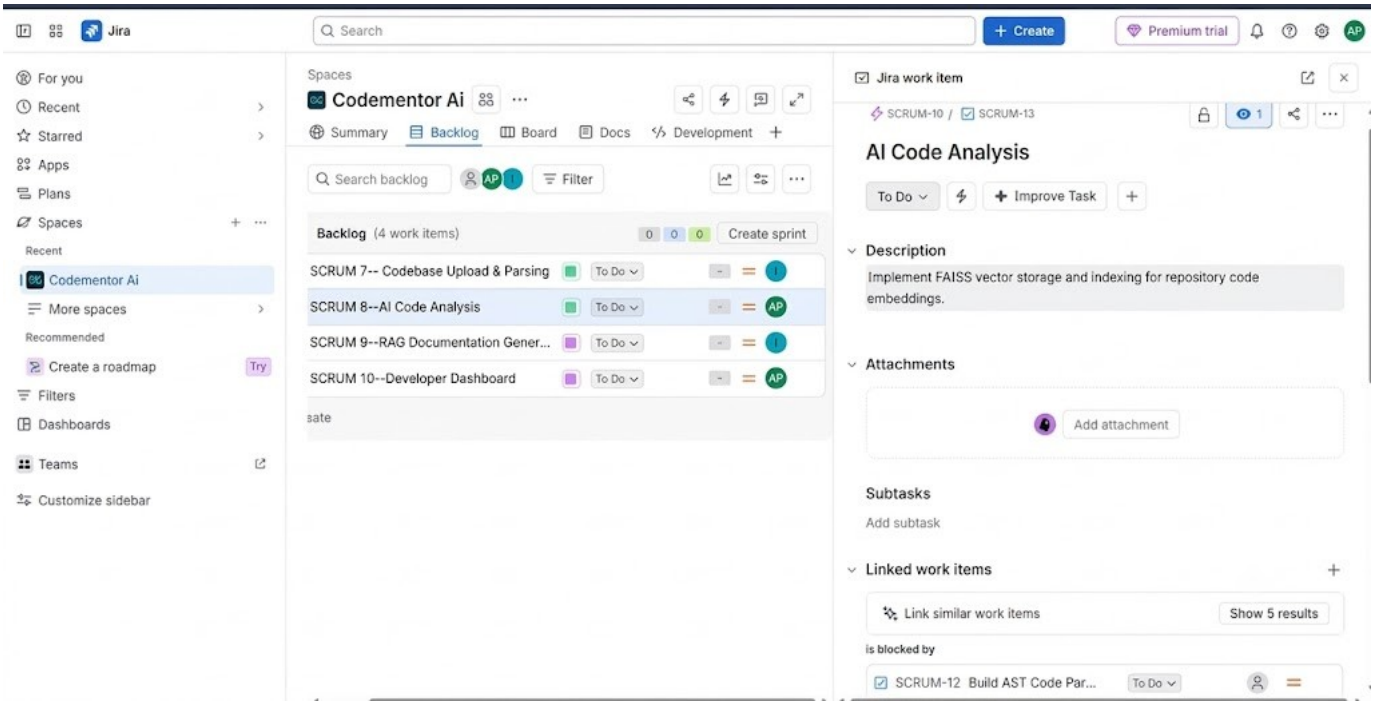
Project Setup: Created a Team-managed Scrum project in Jira Cloud named "CodeMentor AI" and established team member access.



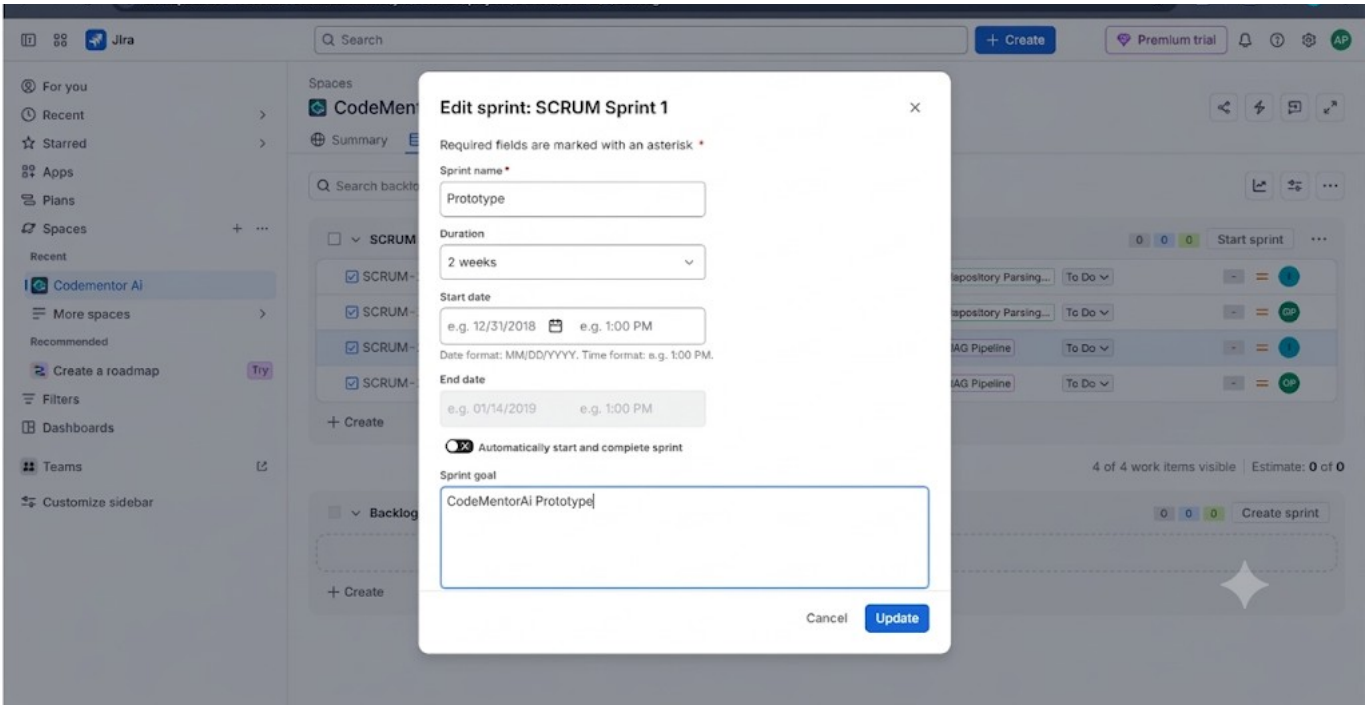
Backlog Refinement & Workload Distribution: Created two Epics (Repository Parsing Engine & RAG Pipeline). Populated the Product Backlog with corresponding user stories and technical tasks, ensuring all items were evenly assigned between Aman and Nikhil.

Dependency Mapping: Configured issue linking (setting a task to be "blocked by" another) to map architectural dependencies.

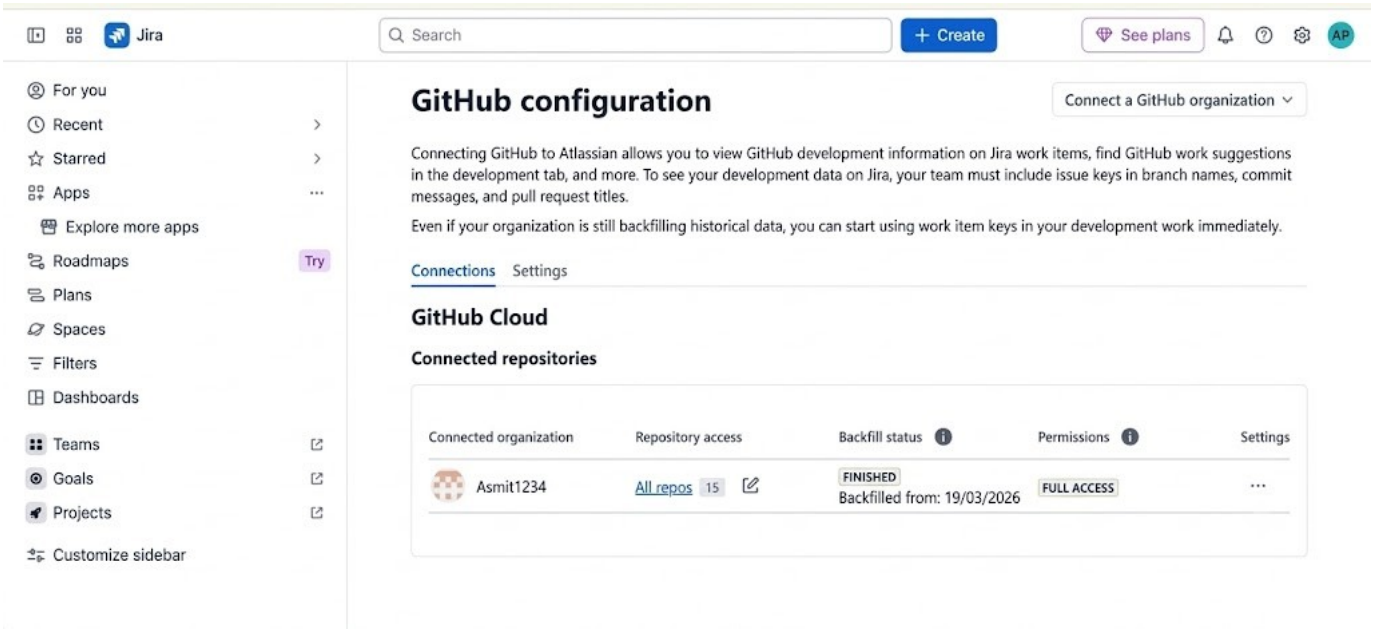




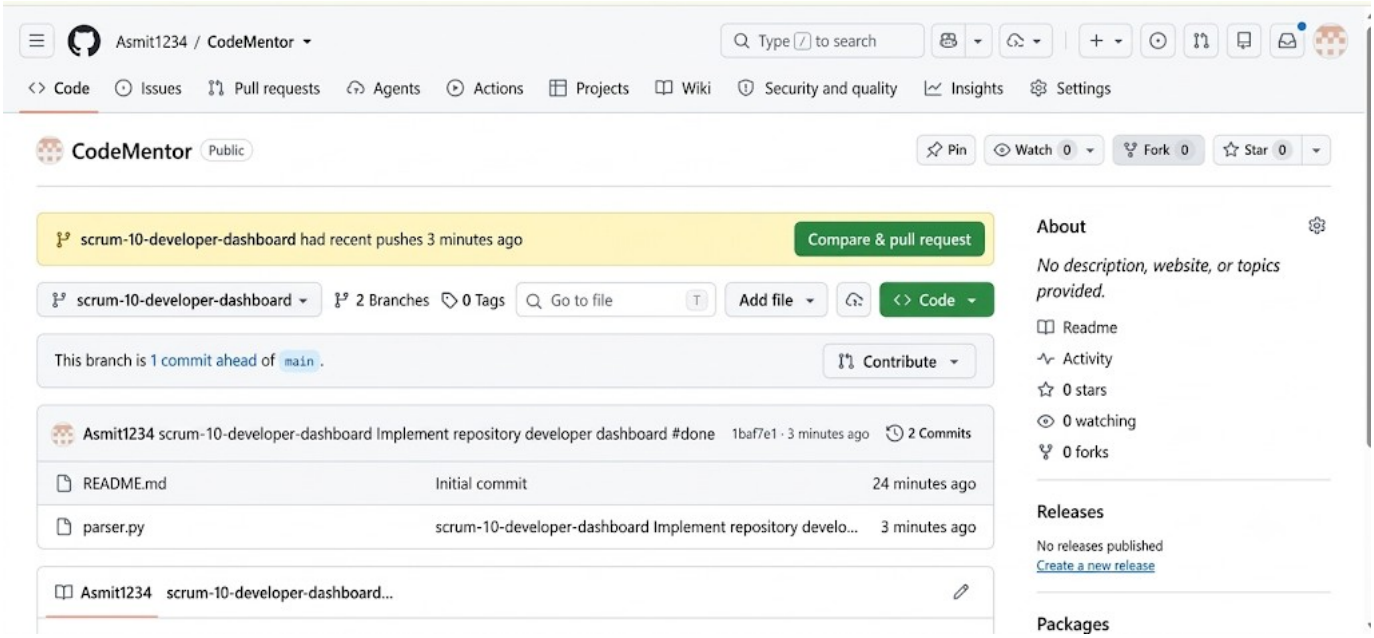
Sprint Execution: Formulated a Sprint goal, loaded the sprint backlog, initialized a 2-week active sprint, and customized the agile board by adding a "Review" workflow column.

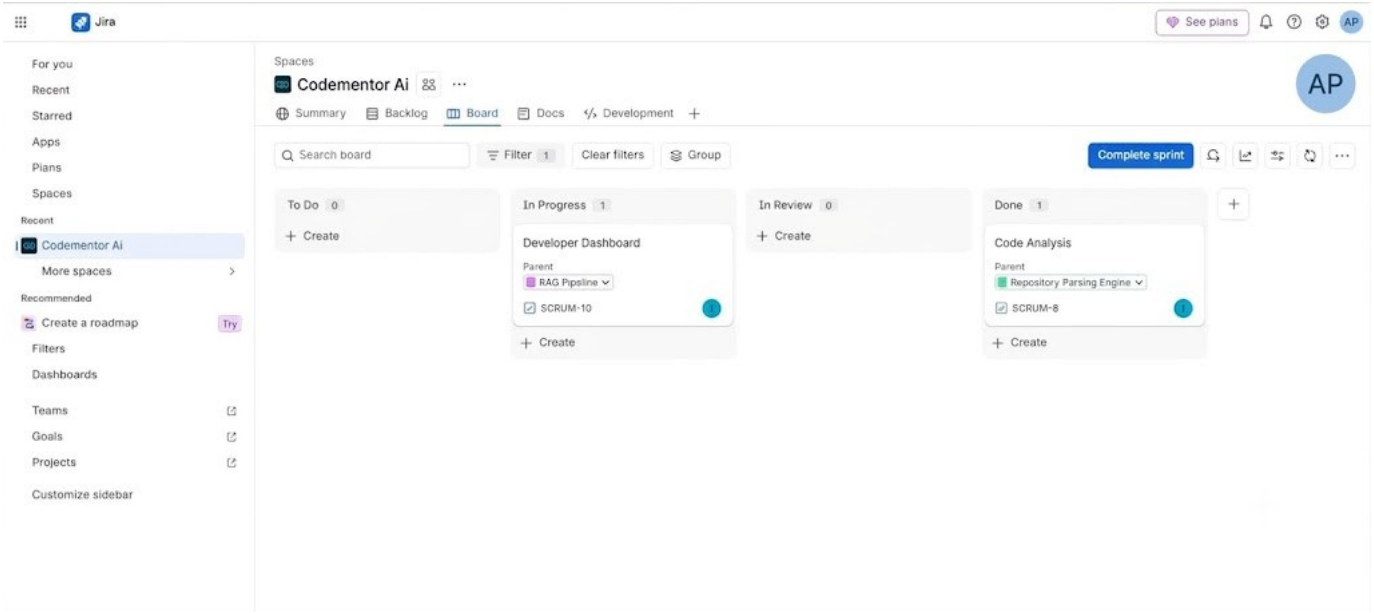


GitHub Integration: Installed the "GitHub for Jira" application and authorized OAuth linkage to the target CodeMentor Ai repository.

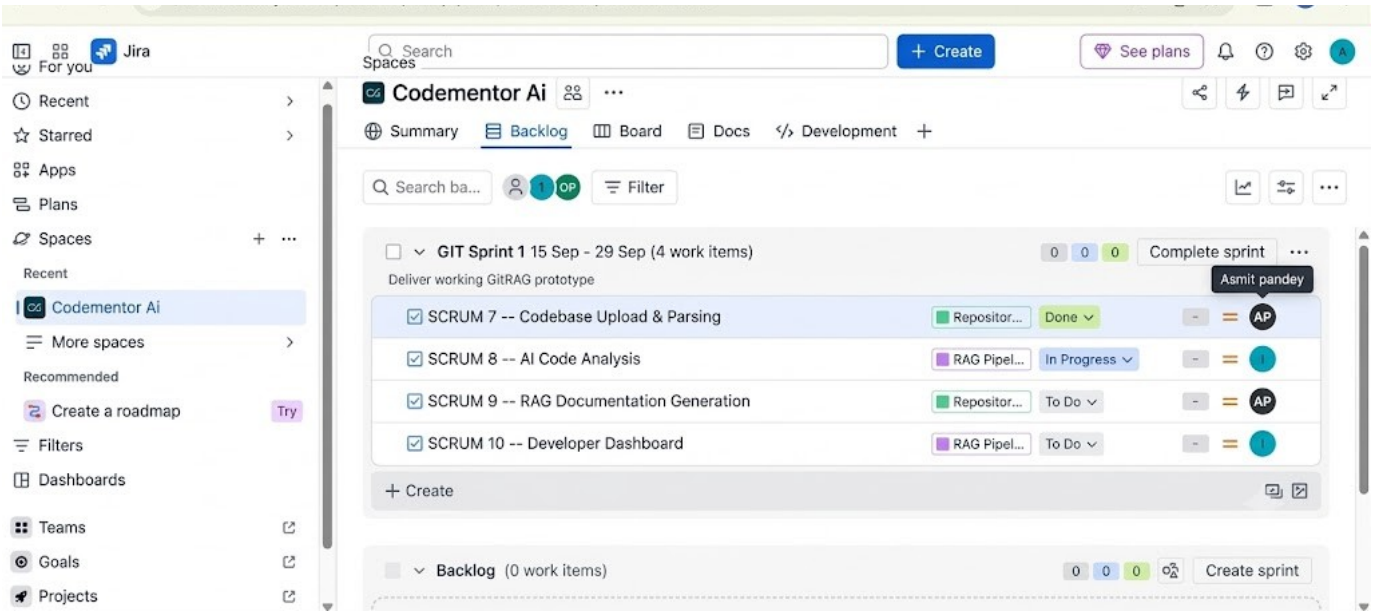


Pipeline Automation: Committed code components to GitHub utilizing Smart Commit syntax (e.g., GIT-4 Dashboard Developer #done).





Verification: Validated that the Jira issues successfully attached the Git commit history directly into the Jira issue development panel, establishing a clear audit trail of who completed which task.



Conclusion:

The experiment successfully demonstrated the Agile lifecycle. By utilizing Jira alongside GitHub, the team effectively managed workload distribution, tracked sprint velocity, and achieved automated DevOps status synchronization, proving clear individual contributions and reducing manual project tracking overhead.